

CST8506 - Lab 1

Dimensionality Reduction – PCA

Due Date: Check Brightspace for due dates.

Introduction

The goal of this lab is to reduce the dimensionality using **PCA** on Diabetes dataset.

You are required to complete this assignment using a Google Colab notebook. Your notebook must be organized into clearly labeled cells, following the steps outlined below. Each step must appear in a separate cell, with an appropriate heading and brief description explaining the purpose of the code and the results of that cell. **If there is no explanation, no marks will be given.** After finishing the lab, take screenshots of each step and paste it in the given answer document and save the document as Lab1_<firstname>.doc.

Before starting the lab, create a diabetes.csv file (add column headers too) from diabetes.arff file that is in the Weka's data folder. There is a diabetes file in sklearn, but the feature names are not clear. So, we will use the diabetes file in Weka that has a meaningful explanation for the features.

Steps (all these steps should be done in Python):

1. Load the csv file using Pandas csv read functionality.
2. Print the attribute names, number of instances, number of attributes and the first 5 instances.
3. Split the dataset into train and test sets.
4. Standardize data in the dataset (fit on the train set and transform it, also, transform the test set).
5. Fit a Random Forest model on the train dataset and then predict the class for the test set. Record the confusion matrix and the accuracy.
6. Apply PCA to find the principal components (fit and transform the train set and transform the test set). (Generate all possible components at this step). Explain the parameter to set the required number of components.
7. Print the explained variance ratios and cumulative explained variance ratios.
8. Generate both scree plots (with explained variance and with cumulative variance) to find the best number of principal components (denote it as d).
9. Once you find d, re-run PCA with d components.
10. Apply Random Forest to the new d-dimensional dataset and find the confusion matrix and the accuracy.
11. Print Confusion matrix before and after applying PCA.
12. Print accuracies before and after applying PCA.
13. Plot the first 2 principal components color coded by each class.
14. Plot the first 3 principal components color coded by each class.

To get grades,

1. You should demo your Python code and results.
2. Submit your answer document and the google colab runnable code in Brightspace. You must use the given answer template to write your answers.
3. **DO NOT ZIP** your files. Zipped files will not be graded.